

investigation. The software and firmware are completely developed in-house, while some of the hardware related jobs are outsourced. Essentially, it's an integrated safety measures solution that attempts to address different causes of accidents.

In line with the concept "not one size fits all," the solution is designed to be highly customisable. There is a range of products and services offered, depending on the solution chosen by the customer.

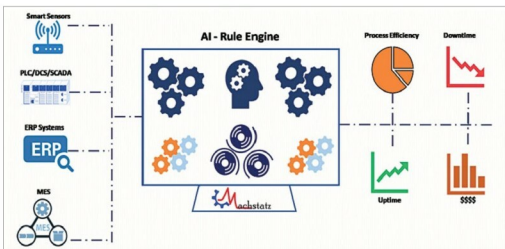
The sensor data collected from the BlackBox is communicated to the cloud server by means of mobile communication. At the cloud end, data analytic algorithms bring out an actionable insight reports for fleet owners.

While vehicle fleet owners are the direct beneficiaries of Cyrrup's solution, other stakeholders include banks, insurance companies and local government bodies.

Currently, the four-wheeler B2B domain comprises individual solution providers and companies that import from China and sell the products as it is. The challenges being faced in the industry are:

1. Companies with big market share offer solutions at high prices
2. The solutions are not easily customisable for individual users
3. The solutions cannot be easily extended to accommodate new client requirements. For each new solution, a new vendor has to be contacted or a new device has to be ordered
4. The service support from Chinese partner is not full-fledged
5. Costly solutions for some of the premium requirements are imported from China as well as EU and Canada, which makes market penetration difficult for local vendors

Cyrrup offers an extensible solution, allowing a lot of customisation. All the features are developed in-house, which gives an edge in providing a good and reliable support.



The complete IIoT platform by Machstatz

The company is attempting to reduce the cost of the solution for premium or niche requirements in the market. It is also focusing on adding extra features on top of basic features like GPS tracking to bring out a market-suited product at lower cost.

Cyrrup won the competition at ZF pitch and a ticket to *Consumer Electronics Show (CES)* in Las Vegas—the world's largest electronics show—to showcase their product. It is also a *NASSCOM 10K Startup* and has been recognised by DIPP, central government of India.

IIoT Solutions

Founded by Sameer Patnaik, Srikant Gupta and Pruthviraj Subudhi, Machstatz Business Solutions provides an end-to-end platform in the industrial IIoT field. Their platform can be used to configure smart sensors using Wi-Fi and Lora-based technologies. It can also be used to configure existing production floor machines through drag-and-drop method. The platform is compatible with legacy control systems, sensors and other data sources like ERP and MES. Once machines

are connected, data extracted from these can be used to analyse and discover hidden patterns and correlations, which can then be used to increase production efficiency and machine uptime and predict failures in real time.

Machstatz also has IIoT modules that can be used for industry-specific use-cases. It also installs smart sensors and devices as part of its offering, providing last-mile connectivity to the manufacturing customers. The team uses industrial automation, Big Data, machine learning algorithms, artificial intelligence and augmented reality solutions to cater to the needs of customers.

Within a span of 10 months, Machstatz has won two customers from different industries (automotive and chemical manufacturing). It plans to focus on these industries and scale up. Once the company validates business case in these industries, it plans to reach out to global customers in areas like oil and gas, and food and beverages.

The start-up has been:

1. Recognised by the government of Karnataka as a promising startup
2. Runner-up in *Elevate 100* programme
3. Among top 25 startups in a pan-India business plan contest organised by Headstart Network and UK-based accelerator E-Spark Viridian Ventures
4. Among top 60 applicants for *Economic Times' Power of Ideas 2017*